# SPOJ Mixtures - Problem Statement

**Problem**: $n$ mixtures are lined up, each with color $c_i \in [0, 99]$. A new mixture is formed by combining two adjacent mixtures $a$ and $b$.

Properties of the resulting mixture:

1. Color: $(a+b) \pmod{100}$
2. Smoke: $a \cdot b$ generated instantly.

$$c_i, c_{i+1} \longrightarrow c_i + c_{i+1} \pmod{100}$$

Smoke: $c_i \times c_{i+1}$

**Goal**: Find minimum total smoke to combine all $n$ mixtures into one.

# SPOJ Mixtures - Dynamic Programming State

**Problem**: Given a sequence of color mixtures, mix them two at a time to minimize smoke.

**Analogy**: Matrix Chain Multiplication (MCM)
Similar to multiplying a chain of matrices, we need to decide the optimal order of mixing colors to minimize the total smoke generated.

## DP State Definition

$dp(i,j)$ = min smoke to mix items i through j
$n \to 0$

$1 \quad 30$

$\therefore \underset{5}{\overset{2}{a}+\overset{+}{b}} = \underset{4}{(a+b)} \bmod 100$

- Rule of mixing colors:
  - Resulting color = $(a+b) \bmod 100$
  - Smoke = $a * b$

$-1 = 2$
$a+b = \bmod 100$
$1-1 = 8$
$-\ (a+b) = (a+b) \bmod 100$
$a+b \qquad a \cdot b$

$$dp(i,j) = \min \{ dp(i,k) + dp(k+1,j) + color(i,k) * color(k+1,j) \}$$
$$\text{for } i \leq k < j$$

Base case: $dp(i,i) = 0$

Comment formula,
- on base case "

# SPOJ Mixtures - Recurrence Relation

We use the dynamic programming approach due to solve the prone in problem, we cmorst chemical by mixing the chemicals with minimum smoke. The clear, the same nrawenucing chemicals, giverevors io inorrection in the functions in main formula.

$$dp(i,j) = \min_{i \leq k < j} \left\{ dp(i,k) + dp(k+1,j) + (color_part1 * color_part2) \right\}$$

$$color_part1 = \left( \sum_{m=i}^{k} \right) \bmod 100$$

$$color_part2 = \left( \sum_{m=k+1}^{j} \right) \bmod 100$$

$dp(i,j)$ = minimum smoke from mixing mixtures $i$ to $j$

base cases:

$dp(i,i) = 0$

$dp(i,u) = 1$ mon [-mixtures] mod 100

$dp(i,i) = 0$

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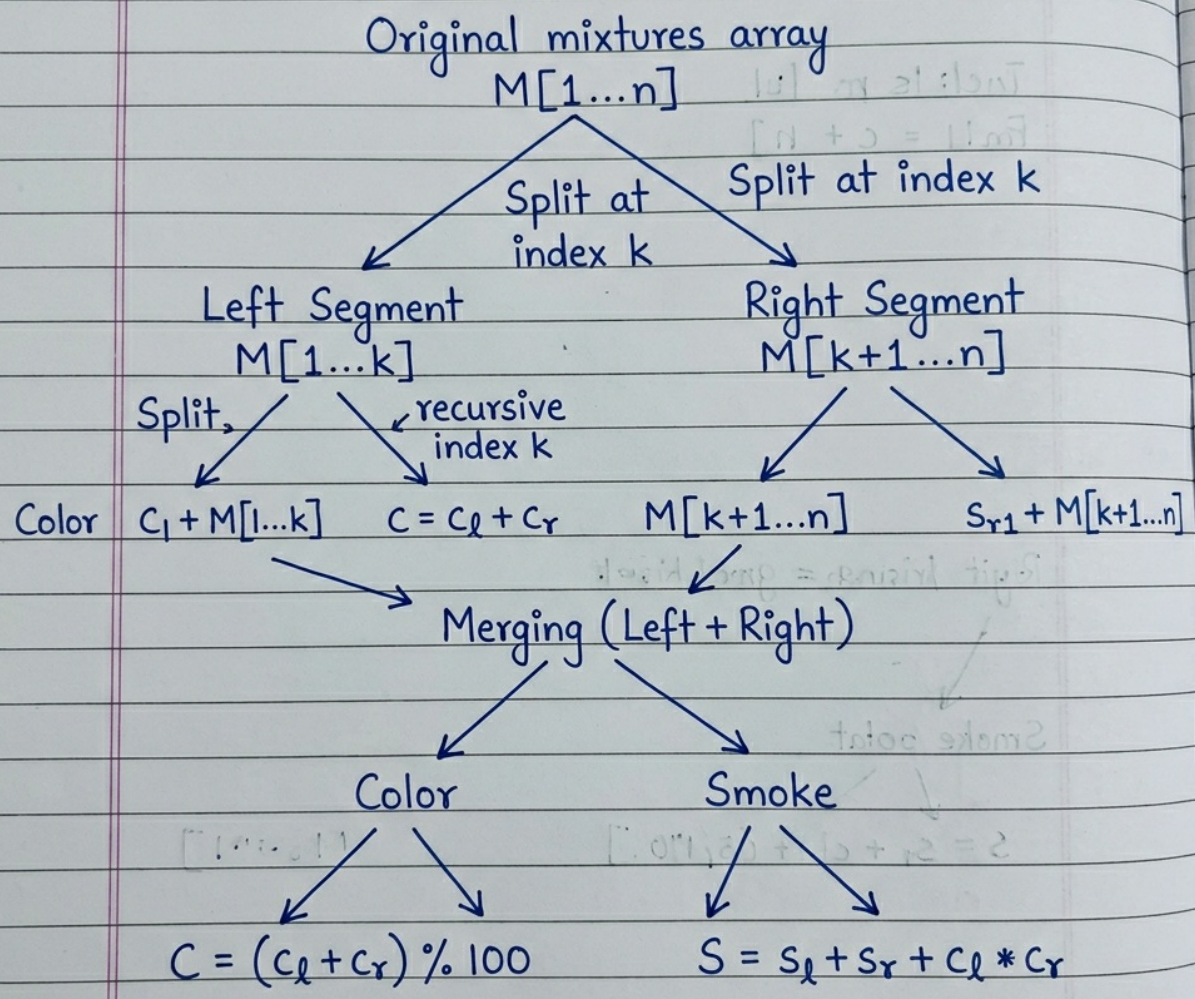
# SPOJ Mixtures - Tree Diagram



# SPOJ Mixtures - Complexity

Time Complexity: $O(n^3)$

Space Complexity: $O(n^2)$

Iterating through length `len`: n iterations (2 to n)

Iterating through start `i`: n iterations (1 to n-len+1)
Iterating through split `k`: n iterations ($i$ to $i$+len-1)
Total is approximately $n \times n \times n = O(n^3)$

a = [2, 3, 5]

dp[i][j] = 2 8
= 1

dp[i][j] = [2, 3, 5]
= 4
= 2